

CSCE 676 DATA MINING · MODULE 11 · LECTURE NOTES

The Pattern You Found

The course's centre: multiplicity, and what survives it.

MODULE OUTCOMES

- Count the hypotheses you actually tested.
- Apply the corrections and explain what each controls.
- Explain why significance and importance diverge at scale.
- Explain validation and pre-registration.
- Defend a found pattern, or abandon it.

1 Counting the hypotheses

EVERY ANALYTIC DECISION YOU MADE WAS A HYPOTHESIS TEST, WHETHER YOU RECORDED IT OR NOT

The explicit statistical tests, obviously — but also every parameter setting you tried, every preprocessing choice you compared, every subgroup you looked at separately, every outlier rule you considered, and every analysis you abandoned because it showed nothing interesting.

Which is exactly why the count cannot be reconstructed afterwards. You do not remember the analyses that failed — they were uninteresting at the time and left no trace — and those are precisely the ones that inflate the multiplicity.

And the garden of forking paths makes it worse still: even a single analysis of a single dataset is a selection, if the analysis you chose depended on what the data looked like — so there is an implicit hypothesis count even when there was no explicit testing at all, which is Gelman and Loken's argument.

So the discipline is to log as you go (Module 01 §4) — and a study whose hypothesis count is unknown cannot be corrected, which means its significance values cannot be interpreted. That is a strong statement and it follows directly from §2's arithmetic.

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RECORD, AS YOU GO
  every test, with its hypothesis and its result
  every parameter value you tried
  every feature set and preprocessing variant
  every subgroup you examined separately
  and every analysis you stopped because it was not
    working

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THE LAST LINE IS THE IMPORTANT ONE. An abandoned analysis is a test whose negative result you discarded, which is exactly what inflates the family.

AND IN AUTOMATED MINING the count is the candidate space the algorithm searched, not the number of results it returned:

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frequent itemsets  -> candidates considered
correlation mining -> pairs tested
model selection    -> configurations evaluated
feature selection   -> features screened

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WHICH IS USUALLY ENORMOUS, and is the honest input to section 2's correction.

For automated mining, the count is the search space rather than the output size — which is the distinction that makes the correction meaningful, and it is the one people get wrong: correcting for the fifty rules your algorithm returned, rather than for the millions of candidates it evaluated, understates the problem by orders of magnitude. **Module 03 §3's point is this one, in the frequent pattern setting.**

2 Corrections

Method	What it controls	Use it when
Bonferroni	The family-wise error rate: the probability of <i>any</i> false positive.	A single false claim would be costly, and m is small. Very conservative for large m .
Holm–Bonferroni	The same family-wise rate, uniformly more powerful.	Always preferable to plain Bonferroni — it is strictly better at no cost in assumptions.
Benjamini–Hochberg	The false discovery rate: the expected fraction of your claims that are false.	You will follow up on many findings — see the note.
Permutation testing	Nothing by itself — it gives you the null distribution empirically.	The test statistic's distribution is unknown, which in this course is most of the time.

The false discovery rate is usually the right target in data mining — you are generating leads for investigation, and tolerating a known fraction of false ones is considerably more useful than guaranteeing none at the cost of finding nothing. Choosing which error rate to control is the real decision, and it follows from what you will do with the findings: a claim that goes into a paper and a lead that goes into a queue deserve different standards.

AND A PERMUTATION TEST IS THE ONE THAT ALWAYS APPLIES

Shuffle the labels, or whatever structure your pattern depends upon, recompute your statistic, and repeat several thousand times — which gives you the null distribution of your statistic on your own data, directly.

So it needs no parametric assumption at all, and it handles test statistics whose distribution nobody has derived — **which includes most of this course's statistics: modularity (Module 07 §3), silhouette (Module 04 §3), rule lift (Module 03), and anomaly scores.**

And it accounts for your data's structure automatically, because **the permuted data retains whatever you held fixed** — the marginals, the degree sequence, the time ordering — which is exactly what makes the null a fair comparison rather than an unrealistic one.

Which makes it the right tool whenever the null comparisons of Modules 04 and 07 are needed — and **the only thing it costs is computation**, which is **the resource this course assumes you have**. There is no good excuse for omitting it.

3 Significance and importance

WITH A MILLION RECORDS, A MEANINGLESS EFFECT IS SIGNIFICANT

The standard error falls as $1/\sqrt{n}$ — so at $n = 10^6$, a correlation of 0.003 has $p < 0.01$ and explains nine millionths of the variance, which is to say nothing at all.

Which means significance testing at this scale answers a question nobody asked: 'is the effect exactly zero' is almost never the question of interest, and at large n the answer is almost always no for any pair of variables with any shared cause whatsoever.

So report the effect size, with a confidence interval — and decide in advance what magnitude would actually matter (Module 01 §4), which is what makes a result interpretable rather than merely publishable.

And the multiplicity problem does not go away when you switch to effect sizes — you still need the correction, applied to the effect sizes rather than to the p-values, because the largest observed effect among many candidates is biased upward by the selection: the winner's curse applies to effect estimates as much as to significance.

4 What actually settles it

- A pattern found and reported. Essentially no evidence, whatever its p-value, because the search that produced it is unaccounted for (§1).
- A pattern found and then corrected for the hypothesis count. Better, and entirely dependent on the count being honest — which is unverifiable by a reader, which is why the next level matters.
- A pattern that survives on held-out data you had not touched. Substantially stronger, because the holdout was not part of the search — so the multiplicity does not apply to it, and a reader can verify that you had one.
- A pattern predicted in advance and then confirmed. Stronger still, because the hypothesis count was one and the prediction is on the record (Module 01 §4's second item).
- And a pattern that predicts data collected afterwards. The strongest available short of an experiment — since no amount of analytic flexibility could have influenced data that did not exist. The jump from the second level to the third is the largest in this list, and it costs only the discipline of splitting the data before you start, which is free.

AND WHAT YOU WOULD NEED TO CALL IT CAUSAL

None of the above establishes causation. A pattern that replicates perfectly on new data can still be a confound — replication establishes that the association is real, not that it is causal, and the distinction is routinely elided.

What would establish it: a randomised experiment, a natural experiment, an instrumental variable, a regression discontinuity, or another credible identification strategy — all of which require something beyond the observational data you have.

So the honest statement names what is specifically missing — 'we would need to randomise exposure to X, which we cannot do because it is determined by Y' — rather than offering a generic 'correlation is not causation' disclaimer, which conveys nothing and is usually a formality.

Which is Project 2's requirement: stating the experiment you would run is the deliverable, because it demonstrates that you know precisely what your evidence does not cover — and that demonstration is worth more than the pattern.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Benjamini & Hochberg — Controlling the False Discovery Rate (free)	§2's third row in the original — and the argument for why the FDR is the right target when you are screening.
Gelman & Loken — The Garden of Forking Paths (free)	§1's hardest point — that an implicit hypothesis count exists even without explicit multiple testing.
Ioannidis — Why Most Published Research Findings Are False (free)	§1 and §4's motivation , with the arithmetic — reread it now that you have the methods.
Sullivan & Feinn — Using Effect Size (free)	§3's recommendation, practically — which effect size measures to use and how to report them.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 Count the hypotheses** in an analysis you have already done, including the abandoned branches.
- 2 Start a hypothesis log for a new analysis** and compare its count against your estimate.
- 3 Compute the candidate space** your frequent itemset mining searched.
- 4 Apply Bonferroni, Holm, and Benjamini-Hochberg** to the same p-values and compare what survives.

- 5 **Explain which error rate each controls** and which you would choose for your task.
- 6 **Run a permutation test** on a modularity score from Module 07.
- 7 **Run one on a silhouette score** from Module 04.
- 8 **Generate 10^6 records of noise** and find a significant correlation; report its effect size.
- 9 **Validate one of your own findings on untouched held-out data** and report the result either way.
- 10 **Write the specific experiment** you would need to call one finding causal.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 What counts as a hypothesis test, and which category is most often omitted?
- 2 Why can the count not be reconstructed afterwards?
- 3 State the garden of forking paths argument.
- 4 In automated mining, what is the count?
- 5 Name four corrections and what each controls.
- 6 Which error rate suits mining, and why?
- 7 Why does a permutation test always apply?
- 8 Why does significance stop being informative at large n ?
- 9 Give the five levels of evidence and the largest jump.
- 10 What would you need to claim causation, and how should you state it?